Naive Bayes Classification Report

Predicting Concert Attendance

Introduction

This activity applies the Naive Bayes classifier to predict whether a person will attend a concert based on several factors.

The attributes used in the dataset are:

- Weather
- Ticket Price
- Distance to Venue
- Friends Going

The target variable is Attend Concert (Yes or No).

The classifier calculates probabilities using Bayes' theorem.

Dataset

	Weather	Ticket Price	Distance to Venue	Friends Going	Attend Concert
0	Sunny	Cheap	Near	True	Yes
1	Rainy	Expensive	Far	False	No
2	Cloudy	Medium	Far	True	Yes
3	Cloudy	Cheap	Near	False	Yes
4	Rainy	Medium	Near	False	Yes
5	Sunny	Expensive	Near	True	No
6	Rainy	Cheap	Far	False	No
7	Sunny	Cheap	Far	True	Yes
8	Cloudy	Expensive	Near	False	No
9	Rainy	Medium	Near	True	Yes
10	Rainy	Expensive	Far	False	No
11	Sunny	Medium	Far	False	Yes
12	Cloudy	Cheap	Near	True	Yes

Total Records = **13**

Prior Probabilities

From the dataset:

- Yes = 8
- No = 5

Therefore:

$$P(Yes) = \frac{8}{13} = 0.615$$
$$P(No) = \frac{5}{13} = 0.385$$

Attend		P(Yes/No)
Yes	8	8/13
No	5	5/13
Total	13	100%

Conditional Probabilities

Weather				
	Yes	No	P(Yes)	P(No)
Sunny	3	1	3/8	1/5
Rainy	2	3	2/8	3/5
Cloudy	3	1	3/8	1/5
Total	8	5	100%	100%

Ticket Price				
	Yes	No	P(Yes)	P(No)
Cheap	4	1	4/8	1/5
Expensive	0	4	0/8	4/5
Medium	4	0	4/8	0/5
Total	8	5	100%	100%

Distance				
	Yes	No	P(Yes)	P(No)
Near	5	2	5/8	2/5
Far	3	3	3/8	3/5
Total	8	5	100%	100%

Friends Going				
	Yes	No	P(Yes)	P(No)
True	5	1	5/8	1/5
False	3	4	3/8	4/5
Total	8	5	100%	100%

Naive Bayes Formula

The probability of a class given the features is calculated using:

$P(y|x1,...,xn) \propto P(y) \cdot \prod i=1nP(xi|y)$

Where:

- Y= class (Yes or No)
- X= input features

The class with the **higher probability** becomes the prediction.

Computation

	Yes	No
Weather: Sunny	= 8/13 (3/8*4/8*5/8*5/8)	= 5/13 (1/5*1/5*2/5*1/5
Ticket Price: Cheap	= 0.615 * 0.073	= 0.385 * 0.0032
Distance: Near	= 0.044	= 0.001
Friends Going: True		

Weather: Rainy	=8/13 (2/8*0/8*3/8*3/8)	= 5/13 (3/5*4/5*3/5*4/5)
Ticket Price: Expensive	= 0.615 * 0	= 0.385 * 0.230
Distance: Far	= 0	= 0.088 ~ 0.09
Friends Going: False		

Weather: Cloudy	= 8/13 (3/8*4/8*3/8*5/8)	= 5/13 (1/5*0/5*3/5*5/8)
Ticket Price: Medium	= 0.615 * 0.044	= 0.385 * 0
Distance: Far	= 0.027	= 0
Friends Going: True		

Weather: Cloudy	= 8/13 (3/8*4/8*5/8*3/8)	= 5/13 (1/5*1/5*2/5*4/5)
Ticket Price: Cheap	= 0.615 * 0.044	= 0.385 * 0.0128
Distance: Near	= 0.02706 ~ 0.03	= 0.004928 ~ 0.005
Friends Going: False		

Weather: Rainy	= 8/13 (2/8*4/8*5/8*3/8)	= 5/13 (3/5*0/5*2/5*4/5)
Ticket Price: Medium	= 0.615 * 0.029	= 0.385 * 0

Distance: Near	= 0.017835 ~ 0.02	= 0
Friends Going: False		

Weather: Sunny	= 8/13 (3/8*0/8*5/8*5/8)	= 5/13 (1/5*4/5*2/5*1/5)
Ticket Price: Expensive	= 0.615 * 0	= 0.385 * 0.0128
Distance: Near	= 0	= 0.004928 ~ 0.005
Friends Going: True		

Weather: Rainy	= 8/13 (2/8*4/8*3/8*3/8)	= 5/13 (3/5*1/5*3/5*4/5)
Ticket Price: Cheap	= 0.615 * 0.018	= 0.385 * 0.0576
Distance: Far	= 0.01107	= 0.022176
Friends Going: False		

Weather: Sunny	= 8/13 (3/8*4/8*3/8*5/8)	= 5/13 (1/5*1/5*3/5*1/5)
Ticket Price: Cheap	= 0.615 * 0.043	= 0.385 * 0.0048
Distance: Far	= 0.026445 ~ 0.03	= 0.001848 ~ 0.002
Friends Going: True		

Weather: Cloudy	= 8/13 (3/8*0/8*5/8*3/8)	= 5/13 (1/5*4/5*2/5*4/5)
Ticket Price: Expensive	= 0.615 * 0	= 0.385 * 0.0512
Distance: Near	= 0	= 0.019712 ~ 0.02
Friends Going: False		

Weather: Rainy	= 8/13 (2/8*4/8*5/8*5/8)	= 5/13 (3/5*0/5*2/5*1/5)
Ticket Price: Medium	= 0.615 * 0.0488	= 0.385 * 0
Distance: Near	= 0.030012	= 0
Friends Going: True		

Weather: Rainy	= 8/13 (2/8* 0/8*3/8*3/8)	= 5/13 (3/5*4/5*3/5*4/5)
Ticket Price: Expensive	= 0.615 * 0	= 0.385 * 0.2304
Distance: Far	= 0	= 0.088704 ~ 0.09
Friends Going: False		

Weather: Sunny	= 8/13 (3/8*4/8*3/8*3/8)	= 5/13 (1/5*0/5*3/5*4/5)
Ticket Price: Medium	= 0.615 * 0.026	= 0.385 * 0
Distance: Far	= 0.01599 ~ 0.02	= 0
Friends Going: False		

Weather: Cloudy	= 8/13 (3/8*4/8*5/8*5/8)	= 5/13 (1/5*1/5*2/5*1/5)
Ticket Price: Cheap	= 0.615 * 0.073	= 0.385 * 0.0032
Distance: Near	= 0.044895	= 0.001232
Friends Going: True		

Predictions for All Records

	Weather	Ticket Price	Distance to Venue	Friends Going	Attend Concert	Predicted
0	Sunny	Cheap	Near	TRUE	Yes	Yes
1	Rainy	Expensive	Far	FALSE	No	No
2	Cloudy	Medium	Far	TRUE	Yes	Yes
3	Cloudy	Cheap	Near	FALSE	Yes	Yes
4	Rainy	Medium	Near	FALSE	Yes	Yes
5	Sunny	Expensive	Near	TRUE	No	No
6	Rainy	Cheap	Far	FALSE	No	No
7	Sunny	Cheap	Far	TRUE	Yes	Yes
8	Cloudy	Expensive	Near	FALSE	No	No
9	Rainy	Medium	Near	TRUE	Yes	Yes
10	Rainy	Expensive	Far	FALSE	No	No
11	Sunny	Medium	Far	FALSE	Yes	Yes
12	Cloudy	Cheap	Near	TRUE	Yes	Yes

Model Accuracy

Correct Predictions = **13**
Total Records = **13**

$$Accuracy = \frac{13}{13} \times 100$$
$$Accuracy = 100\%$$

Conclusion

Using the Naive Bayes classifier, the model successfully predicted concert attendance based on weather, ticket price, venue distance, and whether friends were attending.

The results show that **ticket price and whether friends are going are strong factors influencing the decision to attend the concert**.

The model achieved **100% accuracy on the dataset**, indicating that the probability patterns in the training data strongly match the predictions.

Python Code – Naive Bayes

```
# Hardcoded probabilities from table
prior_yes = 8 / 13
prior_no = 5 / 13

# Conditional probabilities
cond_probs = {
    "Weather": {
        "sunny": {"yes": 3/8, "no": 1/5},
        "rainy": {"yes": 2/8, "no": 3/5},
        "cloudy": {"yes": 3/8, "no": 1/5}
    },
    "TicketPrice": {
        "cheap": {"yes": 4/8, "no": 1/5},
        "expensive": {"yes": 0/8, "no": 4/5},
        "medium": {"yes": 4/8, "no": 0/5}
    },
    "Distance": {
        "near": {"yes": 5/8, "no": 2/5},
        "far": {"yes": 3/8, "no": 3/5}
    },
    "FriendsGoing": {
        "true": {"yes": 5/8, "no": 1/5},
        "false": {"yes": 3/8, "no": 4/5}
    }
}
```

```python
# Get user input
weather = input("Weather (sunny/rainy/cloudy): ").lower()
ticket = input("Ticket price (cheap/medium/expensive): ").lower()
distance = input("Distance to venue (near/far): ").lower()
friends = input("Friends going (true/false): ").lower()

# Compute probabilities
prob_yes = (
    prior_yes *
    cond_probs["Weather"][weather]["yes"] *
    cond_probs["TicketPrice"][ticket]["yes"] *
    cond_probs["Distance"][distance]["yes"] *
    cond_probs["FriendsGoing"][friends]["yes"]
)

prob_no = (
    prior_no *
    cond_probs["Weather"][weather]["no"] *
    cond_probs["TicketPrice"][ticket]["no"] *
    cond_probs["Distance"][distance]["no"] *
    cond_probs["FriendsGoing"][friends]["no"]
)

# Normalize
total = prob_yes + prob_no
yes_percent = (prob_yes / total) * 100
no_percent = (prob_no / total) * 100

# Output
print("\nProbability Results:")
print(f"Attend Concert (YES): {yes_percent:.2f}%")
print(f"Attend Concert (NO): {no_percent:.2f}%")

if yes_percent > no_percent:
    print("Prediction: You are more likely to ATTEND the concert.")
else:
    print("Prediction: You are more likely NOT to attend the concert.")
```

Outputs:

```
PS C:\Users\ASUS> & C:/Users/ASUS/AppData/Local/Programs/Python/Python314/python.exe c:/Users/A
SUS/OneDrive/Desktop/Naive.py
Weather (sunny/rainy/cloudy): rainy
Ticket price (cheap/medium/expensive): cheap
Distance to venue (near/far): near
Friends going (true/false): true

Probability Results:
Attend Concert (YES): 89.06%
Attend Concert (NO): 10.94%
Prediction: You are more likely to ATTEND the concert.
PS C:\Users\ASUS> & C:/Users/ASUS/AppData/Local/Programs/Python/Python314/python.exe c:/Users/A
SUS/OneDrive/Desktop/Naive.py
Weather (sunny/rainy/cloudy): Sunny
Ticket price (cheap/medium/expensive): cheap
Distance to venue (near/far): near
Friends going (true/false): true

Probability Results:
Attend Concert (YES): 97.34%
Attend Concert (NO): 2.66%
Prediction: You are more likely to ATTEND the concert.
PS C:\Users\ASUS>
```

Group – 9

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